

Senior Data Scientist - Take-Home Assignment

Context

You've just joined **Peaks**, a mobile investment app available on iOS and Android. Users download the app, but not all installs lead to a registration. Only a fraction of users who install the app complete registration and become customers. Registered users can then deposit money into managed investment portfolios. The company earns revenue through **management fees on Assets Under Management (AUM)**; currently 0.5% annually.

The company spends aggressively on paid acquisition across **TikTok, Meta, Google App Campaigns, Google Non-Brand Search, and Google Brand Search**. The company also runs a **member-get-member (referral) program**. The CMO has asked you to evaluate the current marketing performance and provide recommendations for the next quarter's budget allocation.

There are several challenges you should be aware of:

- **Privacy changes** have significantly impacted attribution. Many users may be seen as organic installs while they actually come from our marketing channels. Apple's App Tracking Transparency (ATT) framework means most iOS users cannot be attributed to a paid channel. Android tracking has also degraded.
 - **iOS measurement** relies on Apple's SKAdNetwork (SKAN), which provides only aggregated, delayed, and privacy-thresholded conversion data, with no revenue information.
 - The marketing team runs different campaign types, from **brand awareness** to **performance**, but the current measurement treats all channels equally.
 - The company suspects that **not all channels are valued correctly** under the current last-click attribution model.
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Data

You are provided with **5 CSV files** covering approximately 2 years of data.

1. marketing_spend.csv — Daily Ad Spend

Column	Type	Description
date	date	Calendar date
channel	string	TikTok, Meta, Google App Campaigns, Google Non-Brand Search, Google Brand Search, Referral
campaign_id	string	Campaign identifier
campaign_name	string	Campaign name

Column	Type	Description
spend	float	Daily spend (USD)
impressions	int	Ad impressions
clicks	int	Ad clicks

2. app_events.csv — User-Level Attribution Events

Column	Type	Description
user_id	string	Unique user identifier
event_timestamp	datetime	When the event occurred
event_name	string	impression, click, install, registration
platform	string	android or ios
source	string	Attributed channel, Organic, or Referral
campaign_id	string	Campaign ID (null for organic / referral)
country	string	NL, BE, DE

Note: The source field reflects the MMP's last-click attribution. Users who did not consent to tracking appear as Organic even if they originally came from a paid channel.

3. user_profiles.csv — Registered User Demographics

Column	Type	Description
user_id	string	Joins to other tables
age	int	User age (18–65)
gender	string	M or F
registration_date	date	When they signed up
platform	string	android or ios
tracking_enabled	bool	Whether the user consented to ad tracking

Note: This table only contains users who completed registration. Not every install leads to a registration.

4. user_transactions.csv — Deposit & Withdrawal History

Column	Type	Description
user_id	string	Joins to other tables
transaction_timestamp	datetime	When the transaction occurred
transaction_type	string	deposit or withdrawal
amount_usd	float	Transaction amount
running_balance_usd	float	AUM after this transaction
platform	string	android or ios

5. skan_attribution_ios.csv — iOS SKAdNetwork Postbacks

Column	Type	Description
postback_date	date	Date of the SKAN postback
network	string	Channel name
campaign_id	string	Campaign ID
skan_conversion_value	int or null	Conversion value (0–63), null if below privacy threshold
install_count	int	Number of installs in this bucket

Note: SKAN postbacks are delayed and aggregated. The `postback_date` is **not** the install date. Conversion values are integers from 0–63; their meaning is not documented here; part of the challenge is to infer what they represent.

Part 1: Exploratory Analysis & Attribution

The CMO currently evaluates channels using **last-click CPI** (Cost Per Install). She is frustrated that TikTok "looks expensive" despite the brand team insisting it drives growth.

Please investigate:

- Channel Performance** — What does the current last-click attribution tell us about each channel? Calculate CPI (Cost Per Install), CAC (Customer Acquisition Cost), conversion rates, payback period, and any other metrics you find relevant.
- Attribution Gaps** — How does the organic-to-paid ratio differ by platform? Does the `tracking_enabled` field reveal anything about the reliability of attribution? What are the implications for channel measurement?
- Anomaly Detection** — Are there any unusual patterns, time periods, or correlations in the data that suggest the last-click model may be misleading? Look across all available data sources.

4. **Organic Traffic** — Is organic truly organic? Can you identify any relationship between paid spend and organic installs?
 5. **Channel Roles** — Given the campaign data (names, CPMs, CTRs), can you characterize each channel's role in the acquisition funnel? How might this affect how we should measure and value them?
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Part 2: Predictive Modeling

The company needs to understand the **long-term value of a customer** early, ideally within the first few weeks, to make real-time bidding decisions and allocate budget effectively.

1. **Customer Lifetime Value (LTV)** — Using the transaction data, define and compute an appropriate LTV metric for this business model (remember: $\text{revenue} = \text{AUM} \times \text{fee rate}$).
 2. **Early LTV Prediction** — Build a model that predicts a user's long-term value using only information available within the first 14 days after install. What features are most predictive?
 3. **Channel Quality** — Using your LTV model, re-evaluate the channels. Which channel delivers the highest-value users? How does this compare to the CPI-based view? Consider metrics like **payback period** (time to recoup acquisition cost from fee revenue).
 4. **iOS Estimation** — Given the limitations of SKAN data, propose a method to estimate iOS user value.
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Part 3: Strategic Recommendations

Based on your analysis, prepare a brief executive summary for the CMO addressing:

1. **Budget Reallocation** — Should we shift spend between channels? Which channels are overvalued or undervalued?
 2. **Measurement Framework** — How should the company measure marketing performance going forward, given the attribution challenges?
 3. **Quick Wins** — What are 2–3 specific, actionable recommendations the team can implement immediately?
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Deliverables

- **PDF Report** containing your full analysis, visualisations, methodology, and recommendations. This is the **primary deliverable** — it should be detailed, self-contained, and reviewable by both technical and non-technical stakeholders (including the CMO) without any special setup.
- **Executive Summary** (max 1 page, included in the PDF) for the CMO.
- **Jupyter Notebook** (*optional*) — You may include a runnable notebook with your code alongside the PDF, but it does not replace it. The PDF must stand on its own.

Evaluation Criteria

What we're looking for	What we're NOT looking for
Clear analytical thinking and structured approach	Perfect code or production-ready pipelines
Ability to discover non-obvious patterns	Exhaustive analysis of every possible angle
Sound statistical reasoning	Complex ML models for the sake of complexity
Awareness of measurement & privacy challenges	Pixel-perfect visualisations
Honest treatment of uncertainty and limitations	Overconfident conclusions
A well-structured PDF that non-technical stakeholders can follow	A notebook that only a data scientist can run and interpret

We care more about **how you think** than whether you find every pattern. If you run out of time, explain what you would do next.

Good luck! We look forward to reviewing your work.